Innovation Declaration

Our group used large-language-model tools mainly for coding support, debugging, and wording. However, the core design choices came from our own judgement and prior experience in quantitative trading research. We treated the coursework not as a generic machine-learning task, but as a realistic point-in-time trading problem where leakage control, rolling prediction, universe definition, capacity, and transaction costs all matter.

First, we focused heavily on controlling data leakage. This is a common weakness of LLM-generated financial code, because many suggested pipelines accidentally use future information through feature construction, normalization, train-test splitting, or target alignment. We therefore explicitly aligned every modelling row with the information available before portfolio formation, shifted the close-to-open target correctly, and handled earnings timing, short-interest availability, rolling features, and eligibility filters in a point-in-time manner.

Second, we designed a rolling out-of-sample prediction framework. Instead of fitting once on the full sample, each prediction block uses only past data from the same universe. The previous 126 trading days are split into 105 training days and 21 validation days; the selected model is then refit and used to predict the next 21 trading days. This structure reflects how a trading model would actually be researched and deployed, and avoids full-sample look-ahead.

Third, we built a feature-stabilisation pipeline for a noisy set of roughly 250 features. Because many financial factors are highly correlated, we grouped features by Spearman correlation, evaluated cluster-level permutation importance, removed clusters with negative validation importance, and compressed retained clusters using one principal component per cluster. This was intended to reduce noise and make the rolling random forest more stable in short training windows.

Finally, we separated statistical predictability from tradable profitability. We evaluated the model using IC, ICIR, cumulative IC, ablation, capacity, and cost decomposition rather than relying only on final Sharpe. The model achieved positive out-of-sample ranking ability, with IC around 0.02318 and ICIR around 0.1561, but net performance deteriorated after commission, auction slippage, and borrow costs. Our conclusion is therefore that the signal is real, but the close-to-open implementation is structurally expensive because it requires a full daily round trip. This judgement reflects our own trading interpretation, not an off-the-shelf LLM answer.